CONTINUOUS THOUGHT

Continuous Thought Machines

Neural Dynamics and Synchronization as Computation | 2025

Make time part of neural computation: privately parameterized neurons process their own activation histories, while pairwise synchronization becomes the representation used for attention and prediction.

neural dynamics | synchrony | adaptive compute

1 Motivation

RESTORE NEURAL TIMING

- Modern networks reduce neurons to static scalar activations, abstracting away temporal dynamics and synchronization.
- Recurrence usually stores only a shared hidden state rather than a private history per neuron.
- CTM asks whether explicit internal time can support adaptive, interpretable and algorithmic computation.

2 Architecture

AN INTERNAL THOUGHT AXIS

- A self-generated sequence of T internal ticks is decoupled from the input's own dimensions.
- A shared synapse model maps current post-activations plus attended input into new pre-activations.
- Each neuron has a private one-layer MLP that reads its last M pre-activations and emits the next post-activation.

3 Synchronization

TIME BECOMES THE LATENT

- `S_t = Z_t Z_t^T, where Z_t stores neuron activity across internal ticks`
- Sampled neuron-pair correlations are projected into output logits and cross-attention queries.
- Learnable exponential decay lets different pairs emphasize recent or long-range activity while avoiding a full O(D^2) representation.

4 Adaptive learning

OPTIMIZE BEST + CERTAIN

- Each tick produces a prediction, loss and entropy-based certainty score.
- Training averages the loss at the minimum-loss tick and the maximum-certainty tick.
- This allows examples to stop at different ticks without a separately trained halting module.

5 Evidence

DYNAMICS SOLVE TASKS

- A CTM trained on 39 x 39 mazes generalizes to longer routes and 99 x 99 mazes without positional embeddings.
- ImageNet-1K reaches 72.47% top-1 / 89.89% top-5; most examples can halt before 10 of 50 ticks at certainty 0.8.
- On length-64 cumulative parity, 75- and 100-tick runs sometimes reach perfect accuracy and reveal interpretable scanning strategies.

6 Limits & outlook

A NEW, COSTLY AXIS

- Internal sequences extend training time, and private neuron models increase parameter count.
- Experiments are exploratory rather than state-of-the-art and comparisons prioritize breadth over depth.
- Promising directions include language, video, lifelong learning, memory, multimodality and biologically inspired plasticity.